Pricing API for Sound Realty

From listing to closing – faster, smarter price estimates.

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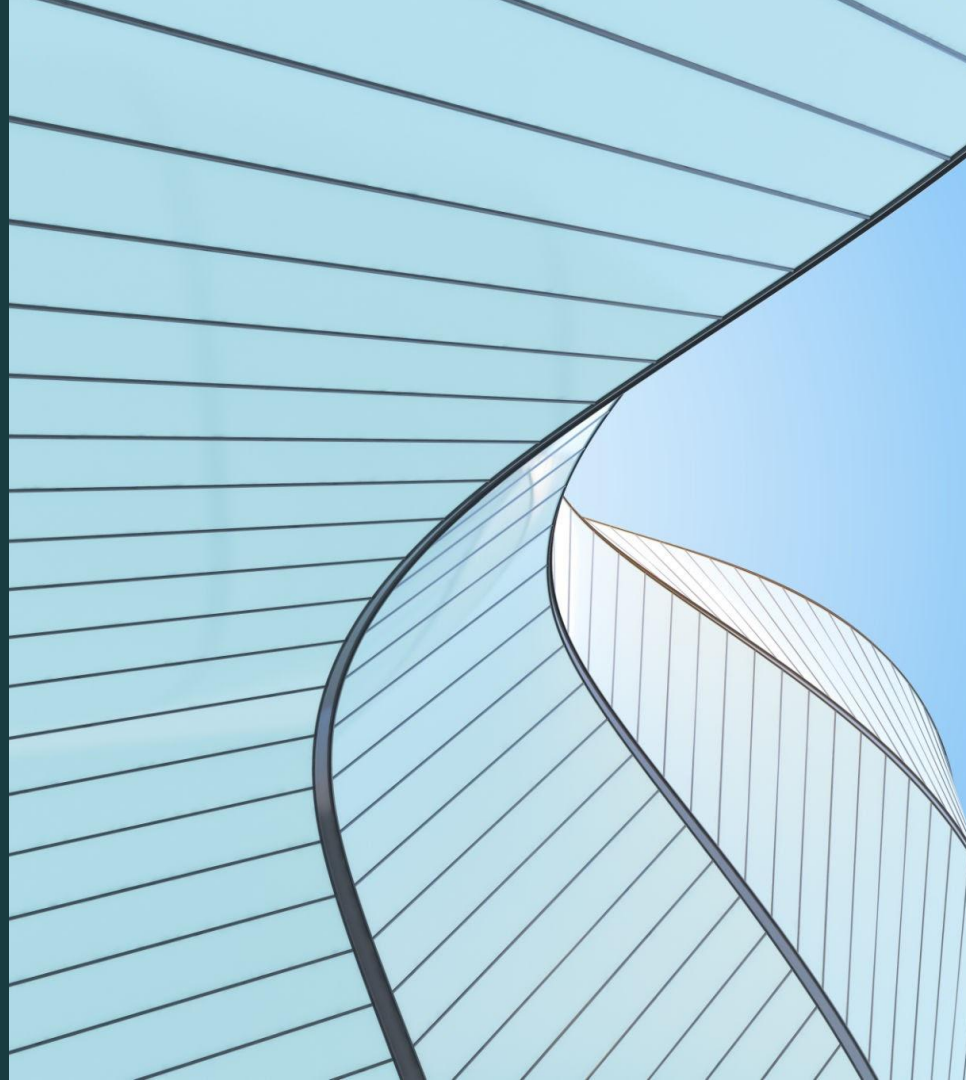

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The Team & The Challenge →

Meet the AI-powered team behind the solution and understand the business problem of manual property valuation.

Our Solution: AI-Powered Pricing API →

Overview of the machine learning model, its rapid prediction capabilities, and the simple input/output structure of the web API.

Live Demonstration →

A real-time showcase of the Pricing API in action, including a health check.

Realty Benefits: Speed & Accuracy →

How faster, more consistent quotes lead to happier sellers and fewer negotiation rounds.

Deployment Roadmap →

Clear steps for integrating the API, selecting model variants, and future improvements.

Key Takeaways & Discussion →

Summary of the solution's impact on efficiency and scalability, followed by a Q&A session.

Our *team*

ML Architect*

Bruno Pedroso

MLE with 10 years of experience in building and maintaining large statistical models

Ideation LLM

Proton Lumo

privacy-first AI assistant created by Proton

Programmer LLM

Claude Haiku 4.5

Cost-efficient AI model, designed for high-speed, agentic, and coding tasks

Presentation and UI LLM

Google Gemini

Gemini can process and reason across text, code, images, video, audio, and voice

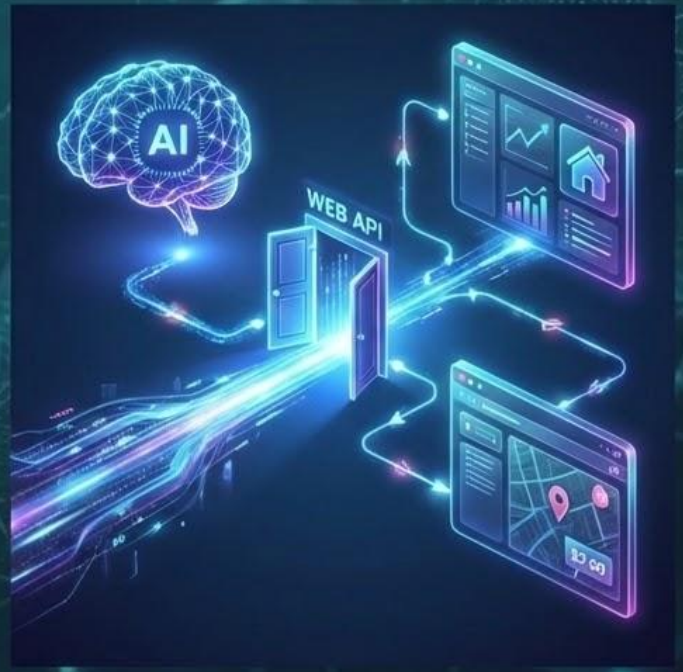
* hopefully

The Business Problem

- Agents spend too many hours manually estimating home values.
- Inaccurate property valuations result in lost deals and time.
- A fast, repeatable, data-driven method is needed for estimates.

Our Solution Overview

- Trained machine learning model predicts property price rapidly.
- Price prediction happens in under one second for fast results
- Solution exposed as a simple, easy-to-use web API
- Integration into agents' existing CRM and web portal workflow



API Concept: Input/Output

- Input is property details, including size, bedrooms, and zip.
- Backend automatically adds relevant demographic data to the input.
- Output provides a fast price estimate with a confidence flag.
- Solution is exposed as a simple, easy-to-use web API

Live Demo: Health Check

Sound Realty Prediction API 1.0.0 OAS 3.1

/openapi.json

REST API for predicting home prices using an ML model

default

GET /health Health Check

POST /predict Predict

POST /predict-minimal Predict Minimal

POST /reload-model Reload Model

Schemas

HTTPValidationError > Expand all object

HealthResponse > Expand all object

PredictionMinimalRequest > Expand all object

Realty Benefits: Faster Deals

- Faster quotes lead to happier sellers for Sound Realty
- Improved model penalizes underpricing more severely
- Consistent property pricing results in fewer negotiation rounds
- The API is scalable and can serve up to 4000 requisitions a minute in the current configuration
- Pricing API provides numerous benefits to Sound Realty

Next Steps for Deployment

Deploy API Container

Deploy the API container on your server or cloud VM.

Select Model Variant

Select the appropriate model variant for your business needs.

Improve Model

Improve model to use more data, from different origins, like property images.

Integrate API

Integrate the pricing API into your existing CRM or web portal.

Add Monitoring

Add model monitoring policies using the ground truth sales as validation.

Key Takeaways and Q&A

- One-click price estimates simplify property valuation
- Pricing data is always up-to-date and highly accurate
- Solution is fully ready to scale with business growth
- Faster, smarter estimates increase agent productivity